

Industrial Internship Report on "Prediction of Agriculture Crop Production and Crop & Weed Detection"

Prepared by

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Executive Summary

This report presents the work carried out during the **6-week Industrial Internship** organized by **upskill Campus** in collaboration with **The IoT Academy** and **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship provided practical exposure to Artificial Intelligence and Machine Learning by allowing me to work on real-world problem statements and develop industry-oriented solutions.

During this internship, I successfully completed two projects: **Prediction of Agriculture Crop Production using Machine Learning** and **Crop & Weed Detection using Deep Learning (YOLO)**. The first project focused on analyzing agricultural datasets to predict crop production based on various factors such as crop type, state, cultivation cost, production cost, and yield. Different data preprocessing techniques and machine learning algorithms were applied to improve prediction accuracy.

The second project involved developing a Crop & Weed Detection system using the YOLO (You Only Look Once) object detection model. The model was trained on an annotated agricultural image dataset to

identify crops and weeds accurately, which can help farmers reduce manual effort and improve precision farming practices.

Throughout the internship, I gained practical experience in Python programming, data preprocessing, data visualization, machine learning, deep learning, Google Colab, GitHub, and project documentation. I also learned the importance of problem-solving, teamwork, and preparing technical reports according to industry standards.

Overall, this internship enhanced my technical knowledge and practical skills in Artificial Intelligence and Machine Learning and provided valuable industry exposure that will support my future academic and professional career.

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1 Preface

Summary of the whole 6 weeks' work:

During the six-week Industrial Internship organized by **upskill Campus** in collaboration with **The IoT Academy** and **UniConverge Technologies Pvt. Ltd. (UCT)**, I worked on two Artificial Intelligence and Machine Learning projects: **Prediction of Agriculture Crop Production using Machine Learning** and **Crop & Weed Detection using Deep Learning (YOLO)**. Throughout the internship, I learned the complete project development lifecycle, including data collection, preprocessing, model development, training, testing, evaluation, documentation, and GitHub repository management. The internship provided valuable practical experience in solving real-world agricultural problems using AI and ML technologies.

About need of relevant Internship in career development:

Industrial internships play a vital role in bridging the gap between theoretical knowledge and practical implementation. They help students develop technical skills, problem-solving abilities, teamwork, and industry exposure, which are essential for building a successful career in Artificial Intelligence and Machine Learning.

Brief about Your project/problem statement:

During this internship, I completed two projects. The first project, **Prediction of Agriculture Crop Production using Machine Learning**, focused on predicting crop production by analyzing agricultural datasets using machine learning techniques. The second project, **Crop & Weed Detection using Deep Learning (YOLO)**, involved developing an object detection model to accurately identify crops and weeds from agricultural images, supporting precision farming and reducing manual effort.

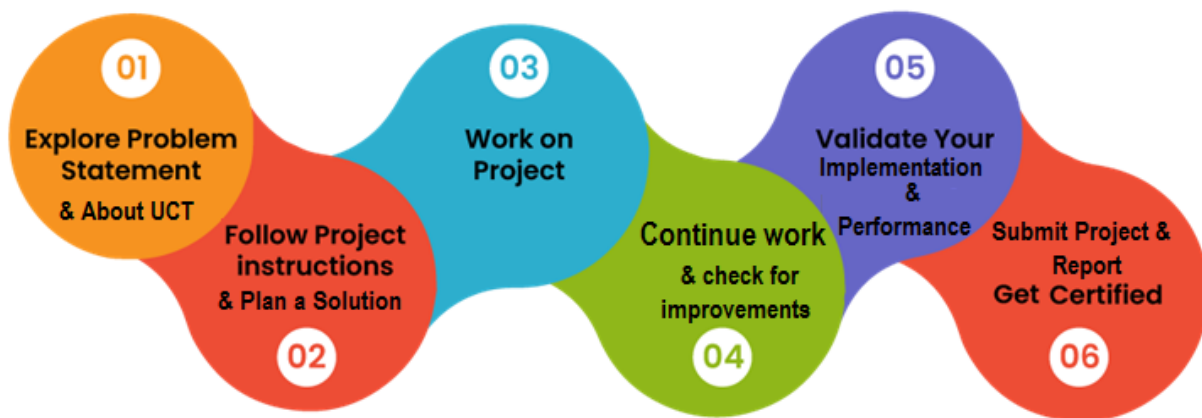
Opportunity given by USC/UCT:

Upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. provided an excellent opportunity to gain practical exposure to industry-oriented AI and Machine Learning projects. The

internship enhanced my technical knowledge and gave me hands-on experience with Python, Machine Learning, Deep Learning, Google Colab, GitHub, and technical documentation.

How Program was planned:

The internship was conducted in a structured manner over six weeks. It began with understanding the assigned problem statements, followed by dataset collection, data preprocessing, model development, training, testing, performance evaluation, project documentation, GitHub repository creation, and final report submission.



Your Learnings and overall experience:

During this internship, I gained practical knowledge of Python programming, data preprocessing, machine learning algorithms, deep learning using YOLO, data visualization, Google Colab, GitHub, and project

documentation. This internship significantly improved my technical skills, analytical thinking, and confidence in developing AI-based solutions for real-world applications.

Acknowledgement:

I sincerely thank **Upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd. (UCT)**, the internship coordinators, mentors, and the faculty members of **Swarna Bharathi Institute of Science and Technology (SBIT)** for their continuous guidance, encouragement, and support throughout this internship. Their valuable suggestions and mentorship helped me successfully complete both projects.

Message to juniors and peers:

I encourage my juniors and fellow students to actively participate in internships and project-based learning opportunities. Such experiences help develop practical skills, improve technical knowledge, and prepare students for future career opportunities in the industry.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



The graphic features the 'uct' logo and the company name 'Uniconverge Technologies' at the top. Below are three service cards: 'IIOT Products' (listing Remote IOs, Wireless IOs, LoRaWAN Sensor Nodes/ Gateways, Signal converter and IoT gateways), 'IIOT Solutions' (listing OEE, Predictive Maintenance, LoRaWAN based Remote Monitoring, IoT Platform, Business Intelligence...), and 'OEM Services' (listing solutions from product design to final production).

uct
Uniconverge Technologies

IIOT Products
We offer product ranging from Remote IOs, Wireless IOs, LoRaWAN Sensor Nodes/ Gateways, Signal converter and IoT gateways

IIOT Solutions
We offer solutions like OEE, Predictive Maintenance, LoRaWAN based Remote Monitoring, IoT Platform, Business Intelligence...

OEM Services
We offer solutions ranging from product design to final production we handle everything for you..

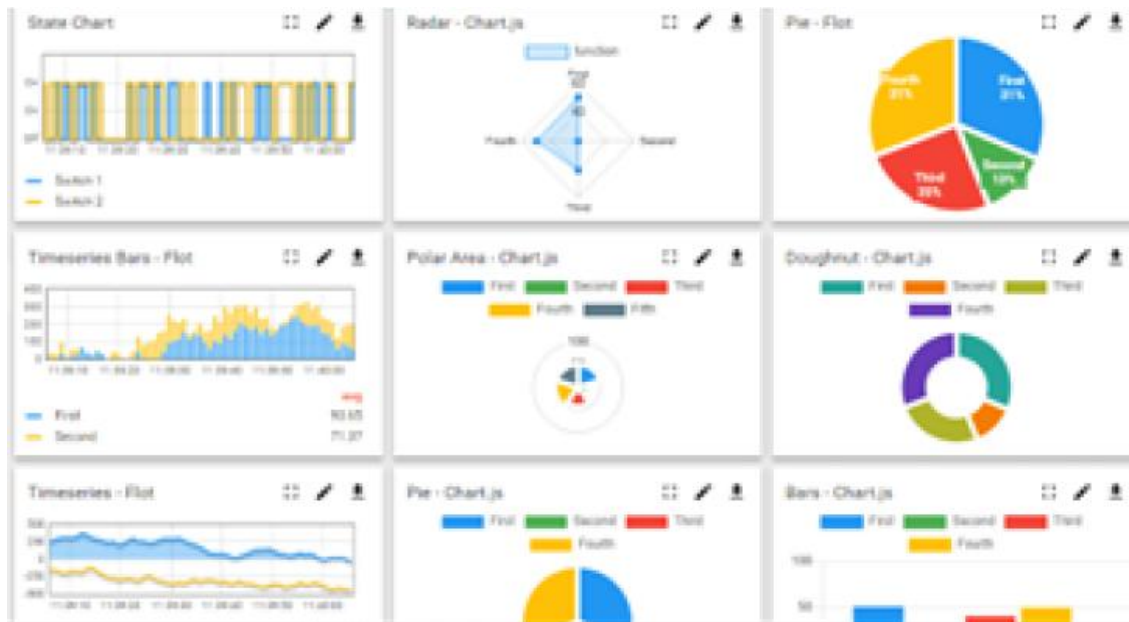
i. UCT IoT Platform ()

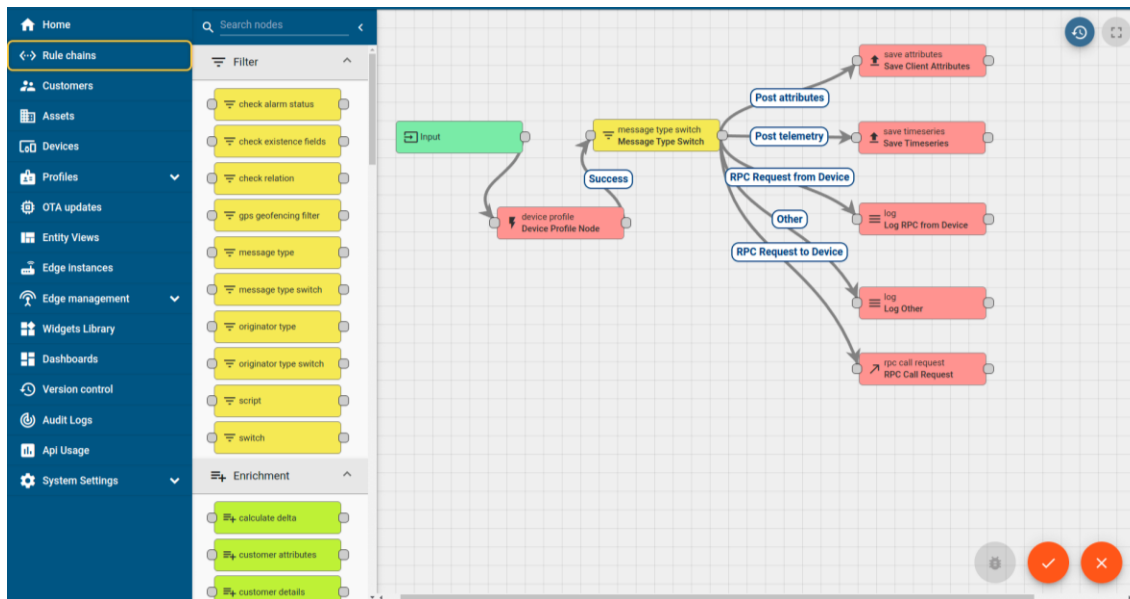
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.

- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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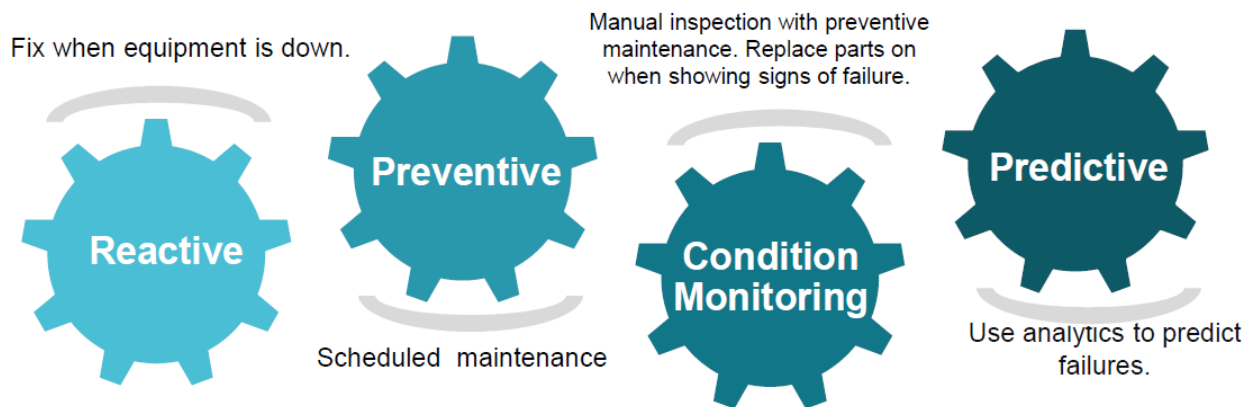


iii. based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

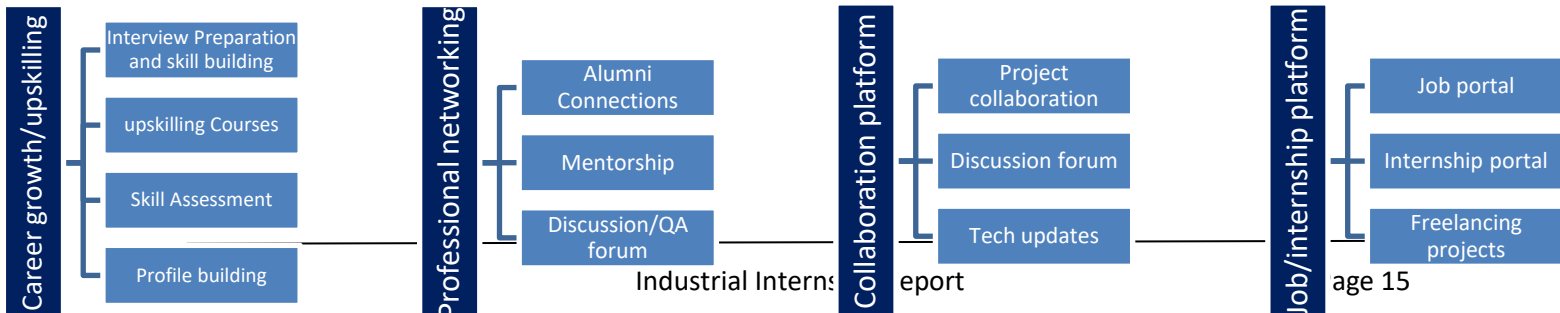
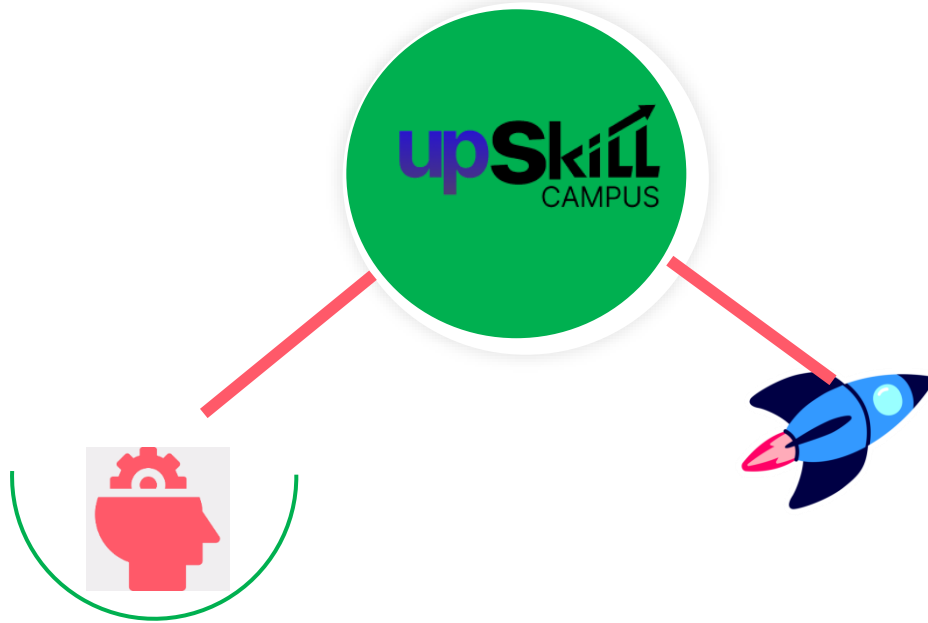
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Python Software Foundation. *Python Documentation*. <https://docs.python.org/3/>
- [2] Scikit-learn Developers. *Scikit-learn Documentation*. <https://scikit-learn.org/stable/>
- [3] Pandas Development Team. *Pandas Documentation*. <https://pandas.pydata.org/docs/>

2.6 Glossary

Terms	Acronym
Artificial Intelligence	AI
Machine Learning	ML
Deep Learning	DL
You Only Look Once	YOLO
Convolutional Neural Network	CNN

3 Problem Statement

Agriculture is one of the most important sectors contributing to food security and economic development. However, farmers face several challenges, including unpredictable crop production due to changing environmental and cultivation factors, as well as the difficulty of identifying weeds at an early stage. These challenges can reduce crop yield, increase production costs, and affect overall agricultural productivity.

The first project, **Prediction of Agriculture Crop Production using Machine Learning**, aims to develop a predictive model that analyzes agricultural data such as crop type, state, cultivation cost, production cost, and yield to estimate future crop production accurately. This helps farmers and agricultural organizations make informed decisions regarding crop planning and resource management.

The second project, **Crop & Weed Detection using Deep Learning (YOLO)**, focuses on developing an object detection system capable of accurately identifying crops and weeds from agricultural images. Early and precise detection of weeds enables farmers to apply targeted weed control methods, reducing unnecessary pesticide usage, minimizing crop damage, and improving farming efficiency.

The objective of these projects is to utilize Artificial Intelligence and Machine Learning techniques to provide smart, efficient, and reliable solutions that support precision agriculture, improve productivity, and assist farmers in making better decisions.

4 Existing and Proposed solution

- **Existing Solutions**

Existing agricultural systems mainly rely on traditional farming methods, manual observations, and historical records to estimate crop production and identify weeds. Crop production prediction is often based on past experiences, making it difficult to accurately estimate future yield under changing environmental conditions. Similarly, weed detection is generally performed through manual field inspection, which is time-consuming, labor-intensive, and prone to human error.

- **Limitations of Existing Solutions**

- Crop production prediction is less accurate due to dependence on manual analysis.
- Difficult to analyze large agricultural datasets efficiently.
- Manual weed identification requires significant time and labor.
- High chances of human error during crop and weed identification.
- Inefficient use of pesticides due to inaccurate weed detection.
- Lack of intelligent decision support for farmers.

- **What is your proposed solution?**

The proposed solution consists of two Artificial Intelligence-based agricultural applications.

The first project, **Prediction of Agriculture Crop Production using Machine Learning**, analyzes agricultural datasets containing crop type, state, cultivation cost, production cost, and yield to predict crop production accurately using Machine Learning algorithms.

The second project, **Crop & Weed Detection using Deep Learning (YOLO)**, uses an object detection model to identify crops and weeds from agricultural images in real time. This helps farmers detect weeds accurately and supports precision farming by reducing manual effort and improving productivity.

Together, these projects provide intelligent, data-driven solutions that enhance agricultural decision-making and improve farming efficiency.

-
- **What value addition are you planning?**
 - Accurate prediction of agricultural crop production using Machine Learning.
 - Automatic detection of crops and weeds using the YOLO Deep Learning model.
 - Reduction in manual effort and time required for crop monitoring.
 - Improved decision-making for farmers based on predictive analytics.
 - Efficient weed management, leading to reduced pesticide usage.
 - Increased agricultural productivity through AI-based smart farming solutions.
 - Better resource utilization and support for precision agriculture.

4.1 Code submission (Github link)

<https://github.com/syedhafsa-cloud/upskillcampus>

4.2 Report submission (Github link) : first make placeholder, copy the link.

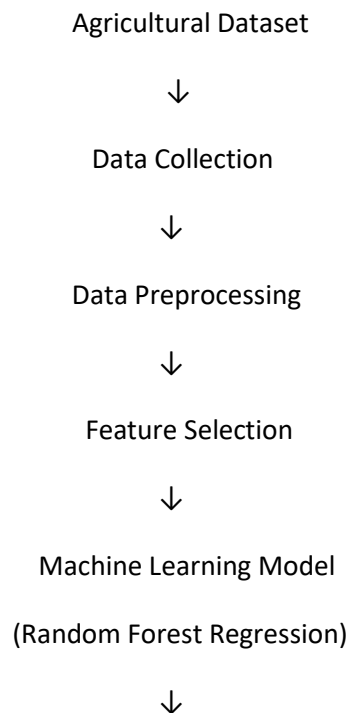
https://github.com/syedhafsa-cloud/upskillcampus/blob/main/Hafsa_Industrial_Internship_Report.pdf

5 Proposed Design/ Model

This internship consists of two Artificial Intelligence and Machine Learning projects. The first project predicts agricultural crop production using historical agricultural data. The second project detects crops and weeds from field images using Computer Vision techniques. Both projects follow a structured workflow involving data collection, preprocessing, model development, testing, and evaluation.

5.1 High Level Diagram (if applicable)

5.1.1 Project 1: Prediction of Agriculture Crop Production:



Crop Yield Prediction



Result Analysis

5.1.2 Project 2: Crop and Weed Detection:

Field Image Dataset



Image Preprocessing



Annotation Processing



YOLO Model Training



Object Detection



Crop / Weed Classification

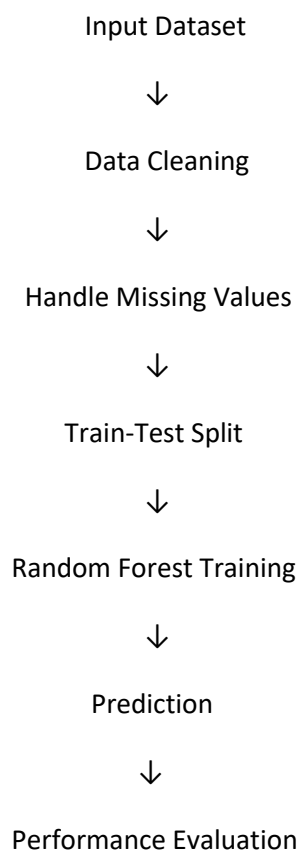


Detection Result

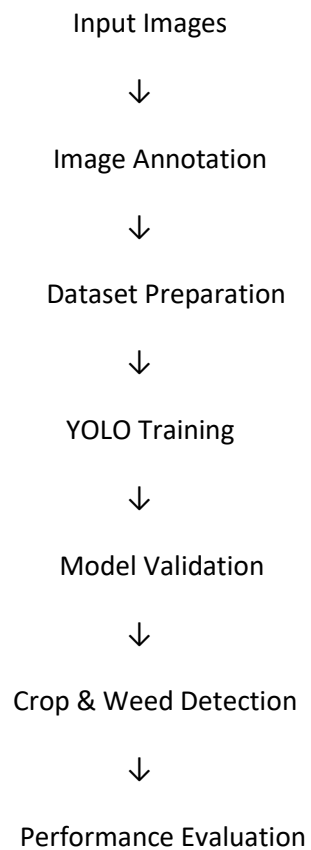
Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.2.1 Project 1: Prediction of Agriculture Crop Production:



5.2.2 Project 2: Crop and Weed Detection:



5.3 Interfaces (if applicable)

The projects were implemented using **Python** in **Google Colab**. Libraries such as **Pandas**, **NumPy**, **Matplotlib**, **Scikit-learn**, **OpenCV**, and **Ultralytics YOLO** were used for data preprocessing, visualization, machine learning model development, object detection, and evaluation.

6 Performance Test

The performance of both projects was evaluated using appropriate datasets. For the crop production prediction project, prediction accuracy and regression metrics were analyzed. For the crop and weed detection project, detection accuracy and object localization performance were evaluated using YOLO.

6.1 Test Plan/ Test Cases

Test Case	Expected Result
Valid agricultural dataset	Accurate crop production prediction
Missing dataset values	Proper preprocessing and prediction
Different crop/state input	Updated production prediction
Valid field image	Correct crop and weed detection
New unseen image	Accurate object detection

6.2 Test Procedure

For the crop production prediction project, the agricultural dataset was cleaned, preprocessed, divided into training and testing datasets, and trained using the Random Forest Regression algorithm. The predicted crop production values were compared with actual values to evaluate performance.

For the crop and weed detection project, annotated field images were prepared and used to train the YOLO model. The trained model was tested on validation images to detect crops and weeds, and the detection results were analyzed for accuracy.

6.3 Performance Outcome

Both projects produced satisfactory results. The crop production prediction model successfully estimated agricultural yield using historical data. The crop and weed detection model effectively identified crops and weeds from field images with good detection performance. These projects demonstrated the practical application of Artificial Intelligence and Machine Learning in modern agriculture.

7 My Learnings

During this internship, I gained practical knowledge in Python programming, data preprocessing, machine learning model development, computer vision, object detection, data visualization, model evaluation, and technical documentation. I also improved my problem-solving skills, analytical thinking, and understanding of real-world AI applications in agriculture.

8 Future work scope

The crop production prediction project can be enhanced by incorporating real-time weather information, soil characteristics, fertilizer usage, and satellite data to improve prediction accuracy.

The crop and weed detection project can be further improved by using larger image datasets, deploying the model on drones or mobile devices for real-time field monitoring, and integrating automated weed removal systems to support precision agriculture.

This content matches the format of your sample report and is tailored specifically to your two internship projects.

